

ePortfolio TM126 – Activity : build-a-switch-and-router-network

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Week :6

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Introduction:

In the part of the TM126 module, I completed an activity involving the construction and configuration of a local area network using cisco Packet Tracer. The objective was to configure two PCs, a switch, and router, ensuring end-to-end connectivity and the correct configuration of IP address according to the provided scenarios.

Objectives

- Configure PCs.
- Configure a router.
- Verify end-to-end connectivity.
- Configure a switch.
- Secure remote access to the router.

Part 1: Procedure and technique steps

- Setting up the network diagram with two PCs (PCA and PCB), a switch, and a router.
- Refer to the addressing table to assign each devices the correct IP address, subnet mask and gateway.

Addressing Table

Device	Interface	IP Address	Subnet Mask	Default Gateway
R1	G0/0/0	192.168.0.1	255.255.255.0	N/A
	G0/0/1	192.168.1.1	255.255.255.0	N/A
S1	VLAN 1	192.168.1.2	255.255.255.0	192.168.1.1
PCA	NIC	192.168.1.3	255.255.255.0	192.168.1.1
PCB	NIC	192.168.0.3	255.255.255.0	192.168.0.1



PCA



S1



R1



PCB

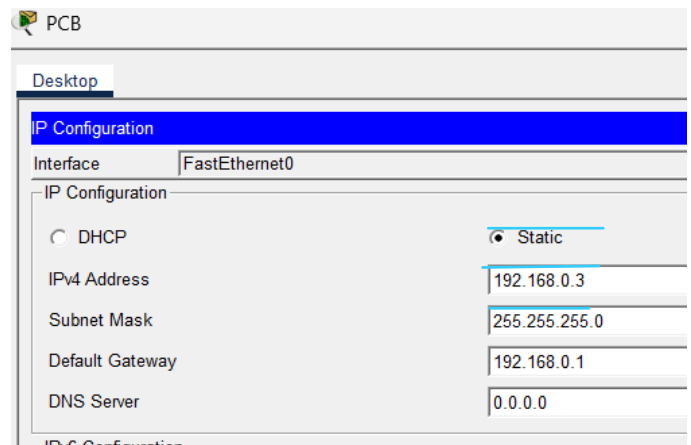
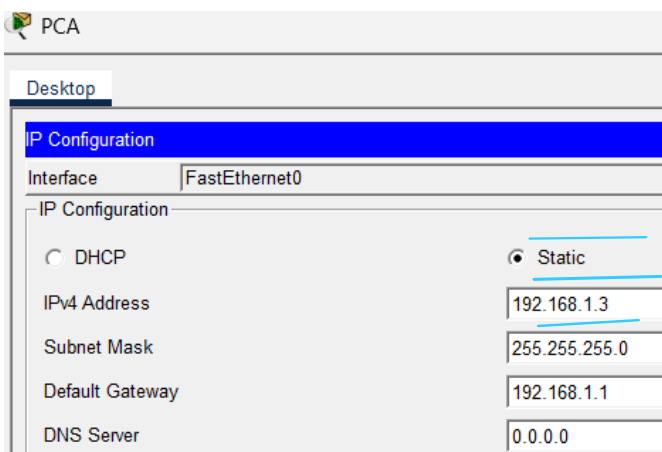
1a) Connecting the equipment : Logical cabling followed by physical connectivity testing

- Connect R1 G0/0/1 to any port on S1.
- Connect PCA to any port on S1.
- Connect PCB to R1 G0/0/0.

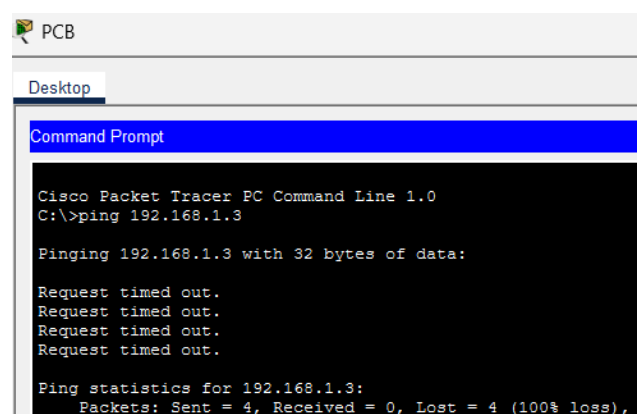
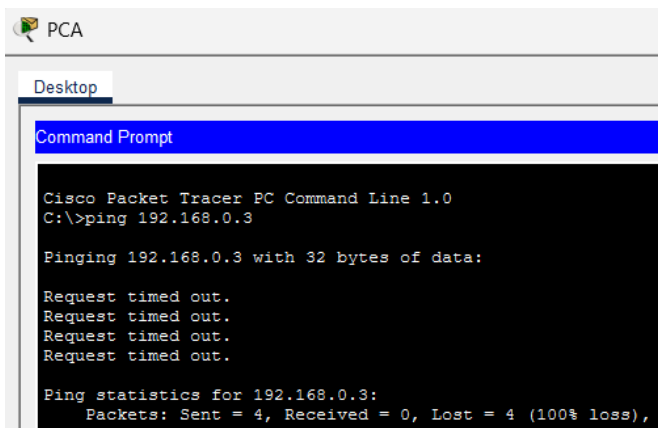


1b) Configuring IP address on PCA and PCB via the desktop.

- Configure the IPv4 address, subnet mask, and default gateway settings on **PCA**.
- Configure the IPv4 address, subnet mask, and default gateway settings on **PCB**.



1c) Test connectivity between PCA and PCB: Ping between PCA and PCB.



Looks like the ping failed between **PCA** and **PCB**

Question Cisco: Why were the pings not successful?

Response : No The router interfaces, which are the default gateways for each PC, have not been configured yet; therefore, the ping traffic is not being routed between the PCA and PCB networks

In this step, you will use the **CLI** tab for **R1** to configure basic settings. You can connect a **Console** cable and access the CLI from **PCA** or **PCB**, if you wish. However, in this activity, you can also just click **R1** to open it.

Part 2: Router configuration

I use the **CLI** tab for **R1** to configure basic settings.

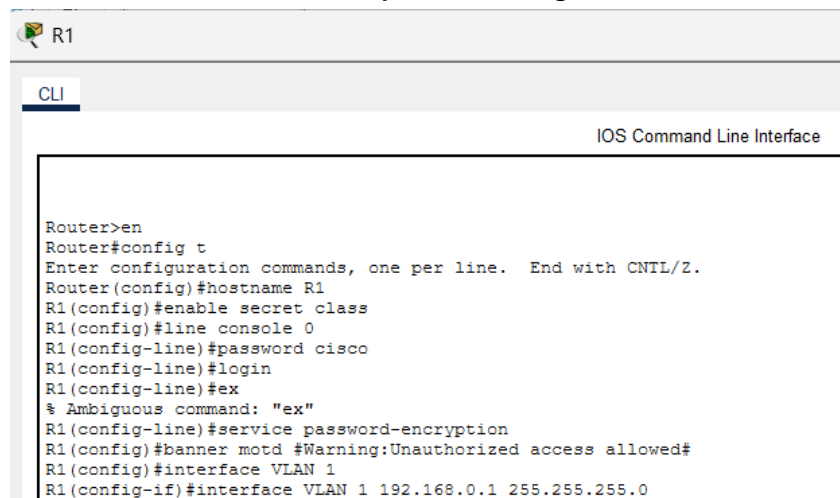
I assign a hostname according to the **Addressing Table**.

I assign **class** as the privileged EXEC encrypted password.

I assign **cisco** as the console password and enable login.

I encrypt the plaintext passwords.

I create a banner that warns anyone accessing the device that unauthorized access is prohibited.



```
R1
CLI
IOS Command Line Interface

Router>en
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname R1
R1(config)#enable secret class
R1(config)#line console 0
R1(config-line)#password cisco
R1(config-line)#login
R1(config-line)#ex
% Ambiguous command: "ex"
R1(config-line)#service password-encryption
R1(config)#banner motd #Warning:Unauthorized access allowed#
R1(config)#interface VLAN 1
R1(config-if)#interface VLAN 1 192.168.0.1 255.255.255.0
```

For G0/0/0, I configure IP addressing according to the Addressing Table and activate the interface.

For G0/0/1, I configure IP addressing according to the Addressing Table and activate the interface.

```
R1#config t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#inter
R1(config)#interface giga
R1(config)#interface gigabitEthernet 0/0/0
R1(config-if)#ip address 192.168.0.1 255.255.255.0
R1(config-if)#no shutdown

R1(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0, changed state to up

R1(config-if)#interface gigabitEthernet 0/0/1
R1(config-if)#ip address 192.168.1.1 255.255.255.0
R1(config-if)#no shutdown

R1(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/1, changed state to up
```

Save the running configuration to the startup configuration file.

```
R1(config-if)#ex
R1(config)#ex
R1#
%SYS-5-CONFIG_I: Configured from console by console

R1#copy
R1#copy run
R1#copy running-config st
R1#copy running-config startup-config
```

2a) verification : Test connectivity between **PCA** and **PCB**

Ping between **PCA** and **PCB**: the command PING to ensure the two are communicating correctly.

PCA

```
C:\>ping 192.168.1.3

Pinging 192.168.1.3 with 32 bytes of data:

Reply from 192.168.1.3: bytes=32 time<1ms TTL=127
Reply from 192.168.1.3: bytes=32 time<1ms TTL=127
Reply from 192.168.1.3: bytes=32 time<1ms TTL=127
Reply from 192.168.1.3: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.1.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

PCB

```
C:\>ping 192.168.0.3

Pinging 192.168.0.3 with 32 bytes of data:

Request timed out.
Reply from 192.168.0.3: bytes=32 time<1ms TTL=127
Reply from 192.168.0.3: bytes=32 time=1ms TTL=127
Reply from 192.168.0.3: bytes=32 time=1ms TTL=127

Ping statistics for 192.168.0.3:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

Question Cisco: Why were the pings between PCA and PCB successful?

Response: Because the router is routing the ping traffic across the two networks. The default settings for the 2960 switch will automatically enable the interfaces that are connected to devices.

the command PING to ensure the two are communicating correctly.

Part 2: Switch configuration

In this step, I use the **CLI** tab for **S1** to configure basic settings.

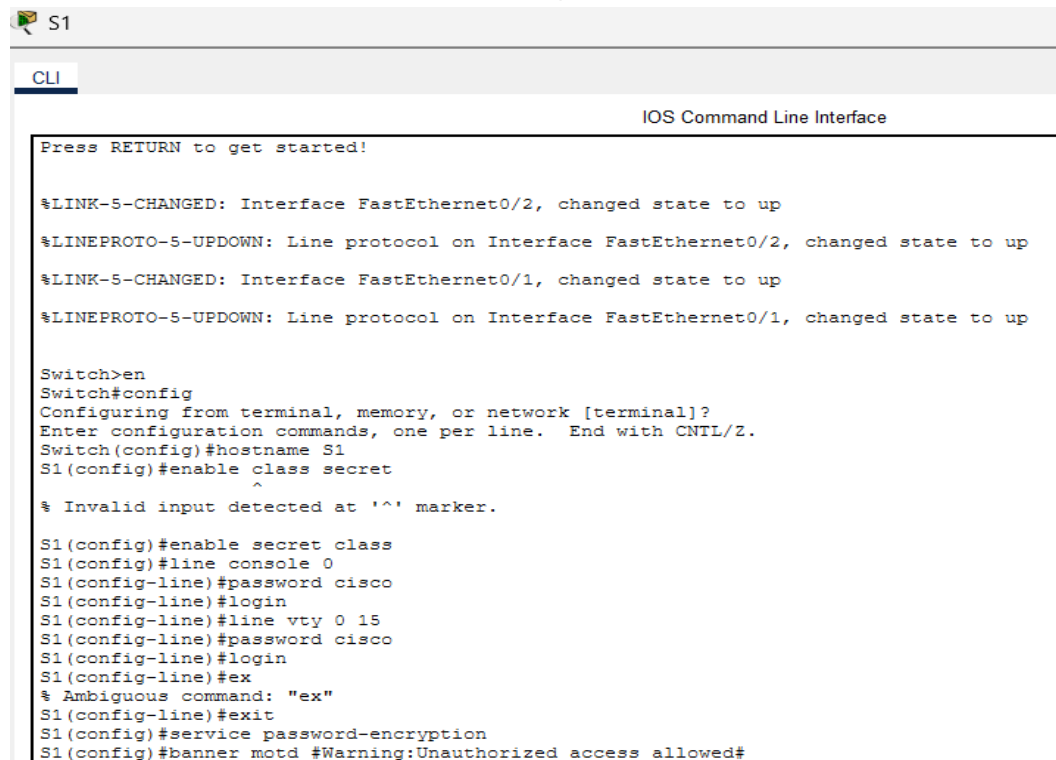
I assign a hostname according to the **Addressing Table**.

I assign **class** as the privileged EXEC encrypted password.

I assign **cisco** as the console password and enable login.

I encrypt the plaintext passwords.

I create a banner that warns anyone accessing the device that unauthorized access is prohibited.



```
S1
CLI
IOS Command Line Interface

Press RETURN to get started!

%LINK-5-CHANGED: Interface FastEthernet0/2, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

Switch>en
Switch#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#hostname S1
S1(config)#enable class secret
% Invalid input detected at '^' marker.

S1(config)#enable secret class
S1(config)#line console 0
S1(config-line)#password cisco
S1(config-line)#login
S1(config-line)#line vty 0 15
S1(config-line)#password cisco
S1(config-line)#login
S1(config-line)#ex
% Ambiguous command: "ex"
S1(config-line)#exit
S1(config)#service password-encryption
S1(config)#banner motd #Warning:Unauthorized access allowed#
```

For VLAN 1, I configure IP addressing according to the Addressing Table and activate the interface.
 Configure the default gateway according to the Addressing Table.
 And I save the running configuration to the startup configuration file.

```
S1(config)#interface VLAN 1
S1(config-if)#ip address 192.168.1.2 255.255.255.0
S1(config-if)#no shutdown

S1(config-if)#
%LINK-3-UPDOWN: Interface Vlan1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan1, changed state to up

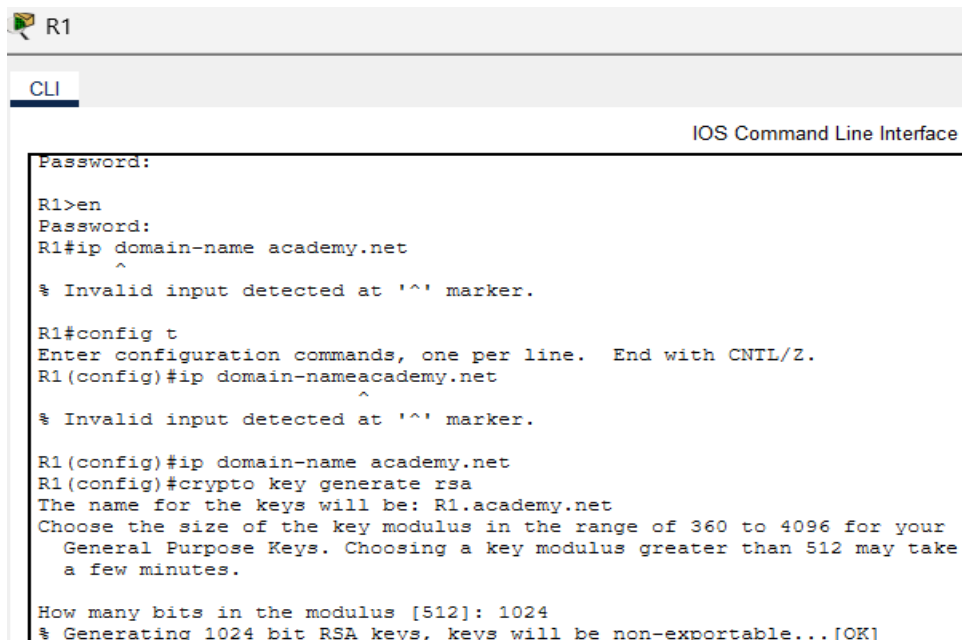
S1(config-if)#ip default
S1(config-if)#ip default
S1(config-if)#ip default g
S1(config-if)#ip default-
S1(config-if)#ip default-gateway 192.168.1.1
S1(config)#exit
S1#
%SYS-5-CONFIG_I: Configured from console by console

S1#copy
S1#copy con
S1#copy run
S1#copy running-config st
S1#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
S1#
```

Part 4: Router configuration

I use the **CLI** tab for **R1** to configure basic settings.

On **R1**, I configure the domain name as **academy.net** and I Generate RSA keys with a **1024** key length.



The screenshot shows the CLI interface for router R1. The user has entered the command 'en' to enter enable mode. Then, they entered 'ip domain-name academy.net' but received an error message: '% Invalid input detected at '^' marker.' This is because the '^' character is not a valid input in this context. The user then entered 'config t' to enter configuration mode. In configuration mode, they entered 'ip domain-name academy.net' again, but received the same error message. Finally, they entered 'crypto key generate rsa'. The system prompted them with 'The name for the keys will be: R1.academy.net' and 'Choose the size of the key modulus in the range of 360 to 4096 for your General Purpose Keys. Choosing a key modulus greater than 512 may take a few minutes.' The user entered '1024' for the modulus size. The system then displayed the message: '% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]'.

```
R1
CLI
IOS Command Line Interface

Password:
R1>en
Password:
R1#ip domain-name academy.net
^
% Invalid input detected at '^' marker.

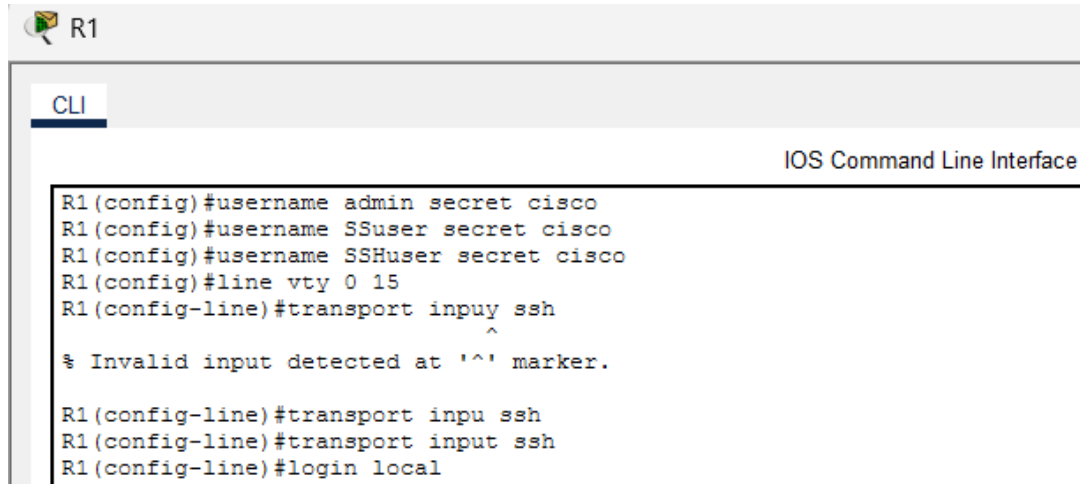
R1#config t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#ip domain-name academy.net
^
% Invalid input detected at '^' marker.

R1(config)#ip domain-name academy.net
R1(config)#crypto key generate rsa
The name for the keys will be: R1.academy.net
Choose the size of the key modulus in the range of 360 to 4096 for your
General Purpose Keys. Choosing a key modulus greater than 512 may take
a few minutes.

How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]
```

I create a user with **SSHuser** as the username and **cisco** as the secret password.

I configure the VTY lines to use the local username database for login credentials. The VTY lines should only allow SSH for remote access.

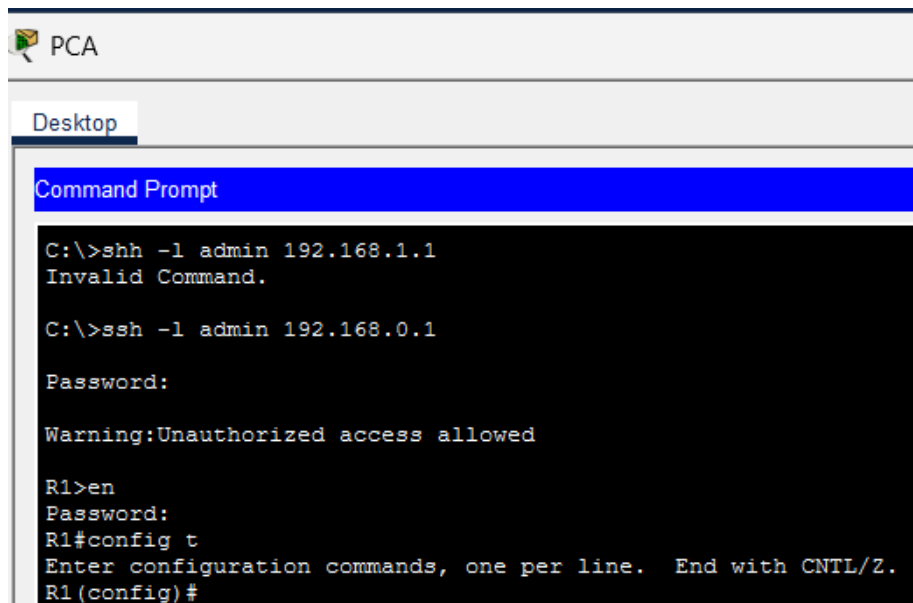
A screenshot of the Cisco IOS Command Line Interface for router R1. The interface shows the configuration of three usernames (admin, SSuser, and SSHuser) with the secret password 'cisco'. It then shows the configuration of VTY lines 0 to 15, setting the transport to 'ssh' and the login to 'local'. There is a typo in the original image where 'input' is misspelled as 'inpu' and 'ssh' is misspelled as 'ssh' (though it looks like 'ssh' in the image, the text says 'inpu ssh' and 'input ssh').

```
R1
CLI
IOS Command Line Interface

R1(config)#username admin secret cisco
R1(config)#username SSuser secret cisco
R1(config)#username SSHuser secret cisco
R1(config)#line vty 0 15
R1(config-line)#transport inpu ssh
^
% Invalid input detected at '^' marker.
R1(config-line)#transport inpu ssh
R1(config-line)#transport input ssh
R1(config-line)#login local
```

4a) **2a) verification :** SSH Remote Access.

From **PCA** or **PCB**, I use the Command Prompt to establish a secure session with R1. At the prompt, use the **ssh** command

A screenshot of the Windows Command Prompt on a PC named PCA. It shows the execution of the 'ssh' command from the C:\ directory. The first attempt to connect to 192.168.1.1 fails with 'Invalid Command.'. The second attempt to connect to 192.168.0.1 is successful, showing a password prompt, a warning about unauthorized access, and then a successful login to the R1 router configuration mode.

```
PCA
Desktop
Command Prompt

C:\>ssh -l admin 192.168.1.1
Invalid Command.

C:\>ssh -l admin 192.168.0.1
Password:
Warning:Unauthorized access allowed
R1>en
Password:
R1#config t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#
```

Question cisco : If the G0/0/1 interface status is administratively down, what interface configuration command would you use to activate the interface?

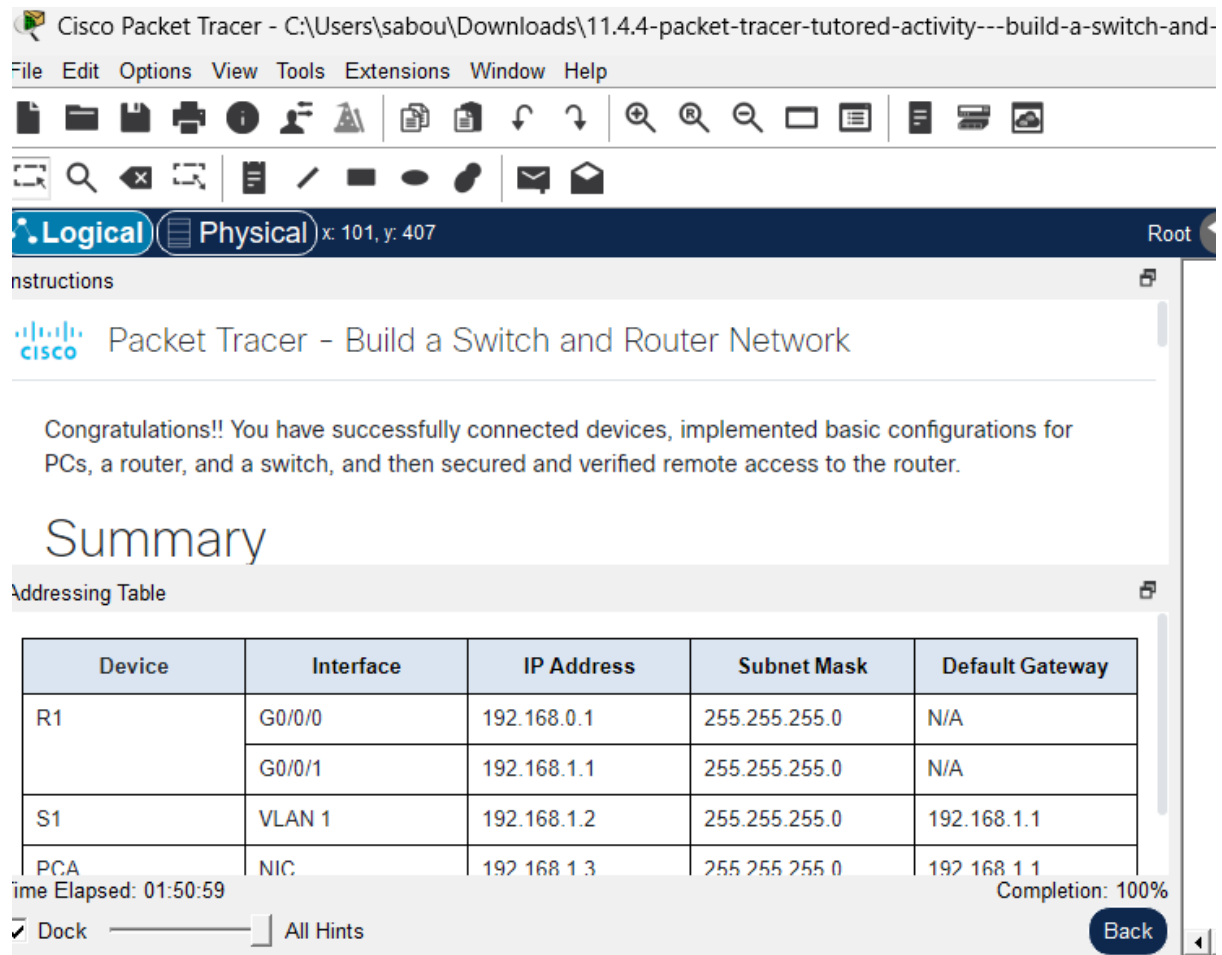
Response: R1(config-if)# no shutdown

Question cisco : What would happen if you had incorrectly configured interface G0/0/1 on the router with an IP address of 192.168.1.2?

Response: PCA would not be able to ping PCB. This is because PCB is on a different network than PCA. A default-gateway router is required to route these packets. PCA is configured to use the IP address of 192.168.1.1 for the default-gateway router, but this address is not assigned to

any device on the LAN. Any packets that need to be sent to the default-gateway for routing will never reach their destination.

Summary: For this activity I achieved a final completion score of (100%). The screenshot below shows the detail assessment results from the activity scoring. The packet tracer file is saved and attached to the assignment.



The screenshot shows the Cisco Packet Tracer interface. The title bar indicates the file path: C:\Users\sabou\Downloads\11.4.4-packet-tracer-tutored-activity---build-a-switch-and-. The menu bar includes File, Edit, Options, View, Tools, Extensions, Window, and Help. The toolbar contains various icons for file operations, network configuration, and simulation. The main window is titled 'Packet Tracer - Build a Switch and Router Network' and displays a congratulatory message: 'Congratulations!! You have successfully connected devices, implemented basic configurations for PCs, a router, and a switch, and then secured and verified remote access to the router.' Below this, a 'Summary' section is visible, followed by an 'Addressing Table'.

Device	Interface	IP Address	Subnet Mask	Default Gateway
R1	G0/0/0	192.168.0.1	255.255.255.0	N/A
	G0/0/1	192.168.1.1	255.255.255.0	N/A
S1	VLAN 1	192.168.1.2	255.255.255.0	192.168.1.1
PCA	NIC	192.168.1.3	255.255.255.0	192.168.1.1

Time Elapsed: 01:50:59 Completion: 100%

☒ Dock ☐ All Hints [Back](#)

Skills acquired:

- Reading and using an IP address table.
- Using Packet Tracer to simulate network scenarios.
- Configuring Cisco routers and switches.
- Troubleshooting connectivity issues.